

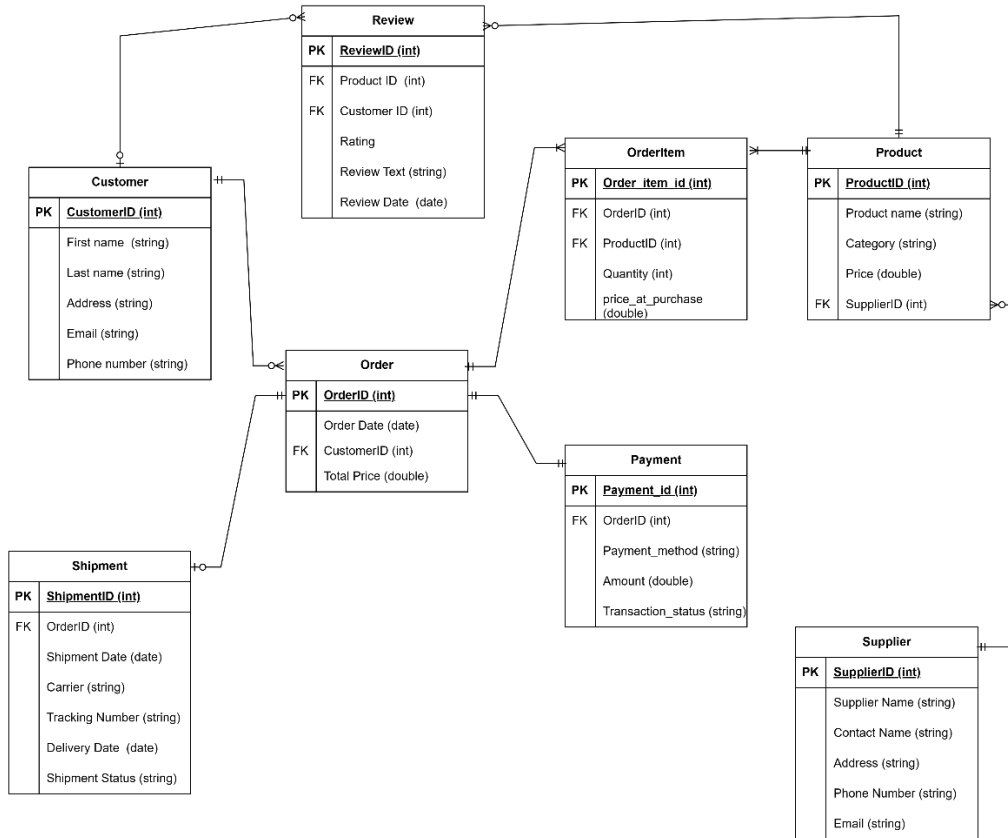


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Data Warehouse Project Documentation

1. Physical model of the source system



2. Fact Tables — Grain & Type

FactSales	
PK	<u>SalesKey</u>
FK	CustomerKey(int)
FK	ProductKey (int)
FK	SupplierKey (int)
FK	OrderDateKey(int)
FK	OrderID(int)
FK	OrderItemID(int)
FK	PaymentKey (int)
	Quantity (int)
	Price_At_Purchase(float)
	Total_amount (float)

FactShipment	
PK	<u>ShipmentFactKey (int)</u>
FK	CustomerKey (int)
FK	ShipmentDateKey(int)
FK	DeliverDateKey(int)
FK	ShipmentKey(int)
FK	OrderID (int)
FK	ShipmentStatusKey(int)
	Delivery_Duration_Days(int)
	ShipmentCount (int)

FactReview	
PK	<u>ReviewFactKey(int)</u>
FK	CustomerKey(int)
FK	ProductKey(int)
FK	ReviewDateKey(int)
FK	ReviewID(int)
	Rating



1.1 Fact_Sales

•**Grain:** Each row in Fact_Sales represents one product line item within one customer order, recorded at the time of purchase.

•**Type:** Transactional

•**Description:** The Fact_Sales table captures every product sold in every order. Since one order can contain multiple products, each product in an order generates a separate row in this fact table. This allows detailed analysis of individual product sales, quantities, and revenue.

1.2 Fact_Shipment

•**Grain:** Each row in Fact_Shipment represents one shipment associated with one order, tracking the shipment from dispatch to delivery.

•**Type:** Accumulating Snapshot

•**Description:** The Fact_Shipment table tracks the lifecycle of each shipment from the moment it is dispatched until it is delivered to the customer. It records both the shipment date and the delivery date, allowing calculation of delivery duration and identification of late deliveries.

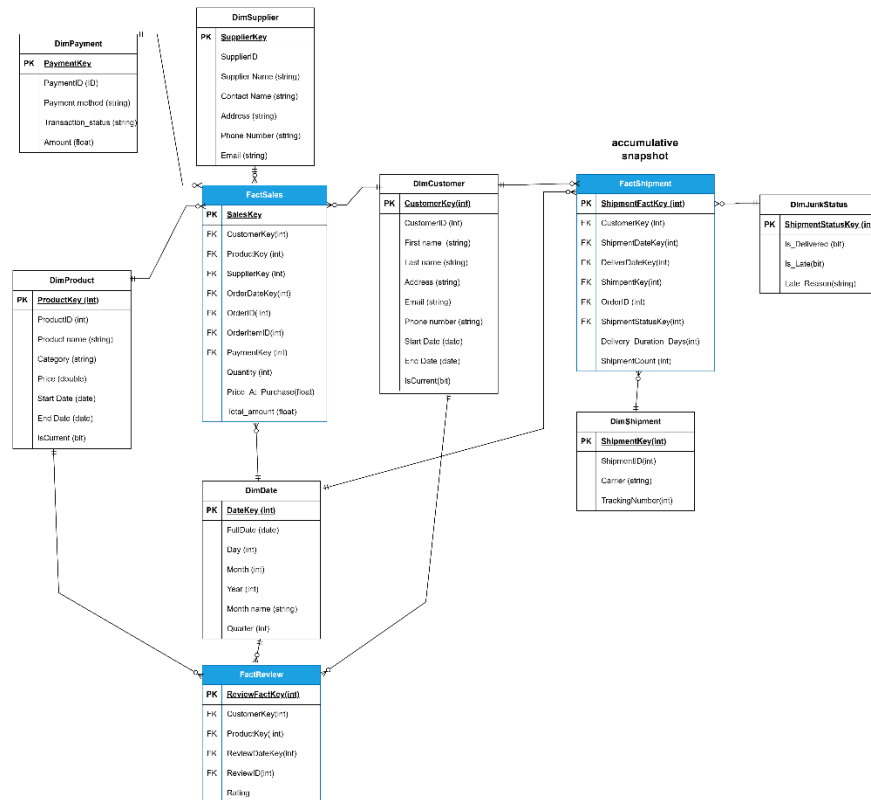
1.3 Fact_Review

•**Grain:** Each row in Fact_Review represents one review written by one customer for one product.

•**Type:** Transactional

•**Description:** The Fact_Review table captures customer feedback for products they have purchased. Each review is recorded as a separate row, containing the rating given by the customer and the date the review was submitted.

2. Dimension Tables — Types & Descriptions



2.1 DimDate — Conformed

•**Description:** The DimDate dimension contains one row for every date between 2020 and 2030. Since dates never change, this dimension uses Type 0, meaning no updates are ever applied. It allows analysis of all facts by day, month, quarter, and year.

Column	Description
DateKey	Surrogate key in format YYYYMMDD
FullDate	Complete date value
Day	Day number of the month
Month	Month number
Year	Year number
Month_Name	Full month name
Quarter	Quarter number (1-4)

2.2 DimCustomer — Type 2 (Track History)



•**Description:** The DimCustomer dimension stores customer information. Since customer details such as address and email can change over time, Type 2 is applied to preserve the full history of changes. Each time a customer's information changes, the old record is expired and a new record is inserted.

Column	Description
CustomerKey	Surrogate key
CustomerID	Natural key from source
First_Name	Customer first name
Last_Name	Customer last name
Address	Customer address
Email	Customer email
Phone_Number	Customer phone
Start_Date	Date record became active
End_Date	Date record was expired
IsCurrent	1 = current, 0 = historical

2.3 DimProduct — Type 2 (Track History)

•**Description:** The DimProduct dimension stores product information. Since product prices and categories can change over time, Type 2 is applied to preserve historical pricing. This ensures that sales records always reference the correct price at the time of purchase.

Column	Description
ProductKey	Surrogate key
ProductID	Natural key from source
Product_Name	Product name
Category	Product category
Price	Product price
Start_Date	Date record became active
End_Date	Date record was expired
IsCurrent	1 = current, 0 = historical

2.4 DimSupplier — Type 1 (Overwrite)



•**Description:** The DimSupplier dimension stores supplier information. Since historical tracking of supplier changes is not required for analysis, Type 1 is applied, meaning any changes to supplier information simply overwrite the existing record.

Column	Description
SupplierKey	Surrogate key
SupplierID	Natural key from source
Supplier_Name	Supplier company name
Contact_Name	Contact person name
Address	Supplier address
Phone_Number	Supplier phone
Email	Supplier email

2.5 DimPayment — Type 1 (Overwrite)

•**Description:** The DimPayment dimension stores payment method and transaction status information. Since historical tracking of payment method changes is not required, Type 1 is applied.

Column	Description
PaymentKey	Surrogate key
OrderID	Natural key from source
Payment_Method	Payment method used
Transaction_Status	Completed, Pending, or Failed

2.6 DimShipment — Type 1 (Overwrite)

•**Description:** The DimShipment dimension stores carrier and tracking information for each shipment. Type 1 is applied since historical tracking of carrier changes is not required.

Column	Description
ShipmentKey	Surrogate key
ShipmentID	Natural key from source
Carrier	Shipping carrier name
Tracking_Number	Shipment tracking number



2.7 DimJunkStatus — Conformed

•**Description:** The DimJunkStatus dimension is a junk dimension that combines shipment status flags into a single dimension table. It contains all possible combinations of Is_Delivered, Is_Late, and Late_Reason. Since these combinations are pre-defined and never change, Type 0 is applied.

Column	Description
ShipmentStatusKey	Surrogate key
Is_Delivered	1 = Delivered, 0 = Not Delivered
Is_Late	1 = Late, 0 = On Time
Late_Reason	Reason for late delivery

3. Analytical Queries and Insights

Query 1 — Total Revenue

```
SELECT
    SUM(f.Total_Amount) AS Total_Revenue
FROM Fact_Sales f
JOIN DimPayment p ON f.PaymentKey = p.PaymentKey
WHERE p.Transaction_Status = 'Completed'
```

Query 2 — Revenue by Month and Year

```
SELECT
    d.Year,
    d.Month_Name,
    SUM(f.Total_Amount) AS Monthly_Revenue,
    COUNT(DISTINCT f.OrderID) AS Total_Orders
FROM Fact_Sales f
JOIN DimDate d ON f.OrderDateKey = d.DateKey
JOIN DimPayment p ON f.PaymentKey = p.PaymentKey
```



```
WHERE p.Transaction_Status = 'Completed'
GROUP BY d.Year, d.Month_Name, d.Month
ORDER BY d.Year, d.Month
```

Query 3 — Revenue by Product

```
SELECT
    p.Product_Name,
    p.Category,
    SUM(f.Total_Amount) AS Revenue,
    SUM(f.Quantity) AS Units_Sold
FROM Fact_Sales f
JOIN DimProduct p ON f.ProductKey = p.ProductKey
JOIN DimPayment py ON f.PaymentKey = py.PaymentKey
WHERE py.Transaction_Status = 'Completed'
AND p.IsCurrent = 1
GROUP BY p.Product_Name, p.Category
ORDER BY Revenue DESC
```

Query 4 — Top 10 Best Selling Products

```
SELECT TOP 10
    p.Product_Name,
    p.Category,
    SUM(f.Quantity) AS Units_Sold,
    SUM(f.Total_Amount) AS Revenue
FROM Fact_Sales f
JOIN DimProduct p ON f.ProductKey = p.ProductKey
WHERE p.IsCurrent = 1
GROUP BY p.Product_Name, p.Category
ORDER BY Units_Sold DESC
```

Query 5 — Average Delivery Time per Carrier

```
SELECT
    s.Carrier,
    AVG(f.Delivery_Duration_Days) AS Avg_Delivery_Days,
    MIN(f.Delivery_Duration_Days) AS Min_Days,
    MAX(f.Delivery_Duration_Days) AS Max_Days,
    SUM(f.Shipment_Count) AS Total_Shipments
FROM Fact_Shipment f
JOIN DimShipment s ON f.ShipmentKey = s.ShipmentKey
GROUP BY s.Carrier
ORDER BY Avg_Delivery_Days ASC
```

Query 6 — Average Product Rating



```
SELECT
    p.Product_Name,
    p.Category,
    AVG(CAST(f.Rating AS FLOAT)) AS Avg_Rating,
    COUNT(*) AS Total_Reviews
FROM Fact_Review f
JOIN DimProduct p ON f.ProductKey = p.ProductKey
WHERE p.IsCurrent = 1
GROUP BY p.Product_Name, p.Category
```

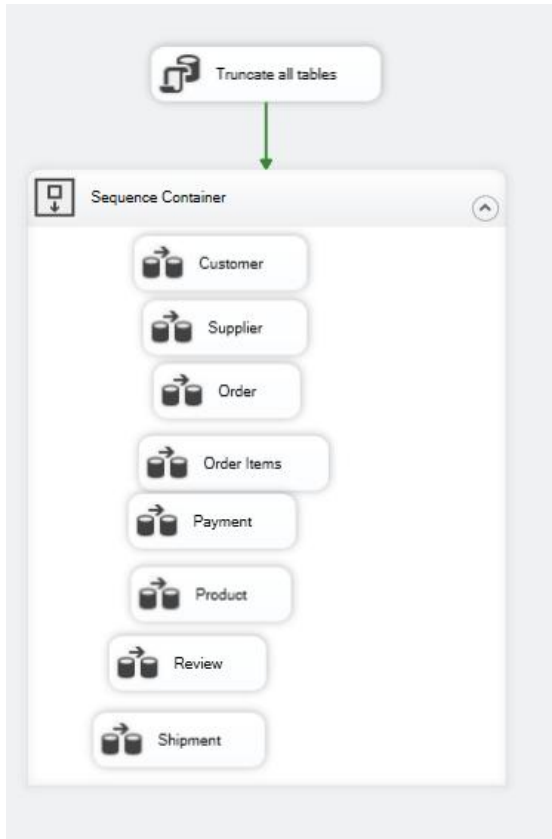
Query 7— Top 10 Best Rated Products

```
SELECT TOP 10
    p.Product_Name,
    p.Category,
    AVG(CAST(f.Rating AS FLOAT)) AS Avg_Rating,
    COUNT(*) AS Total_Reviews
FROM Fact_Review f
JOIN DimProduct p ON f.ProductKey = p.ProductKey
WHERE p.IsCurrent = 1
GROUP BY p.Product_Name, p.Category
HAVING COUNT(*) >= 5
ORDER BY Avg_Rating DESC
```

4. Using SSIS in VS [Staging Area]:

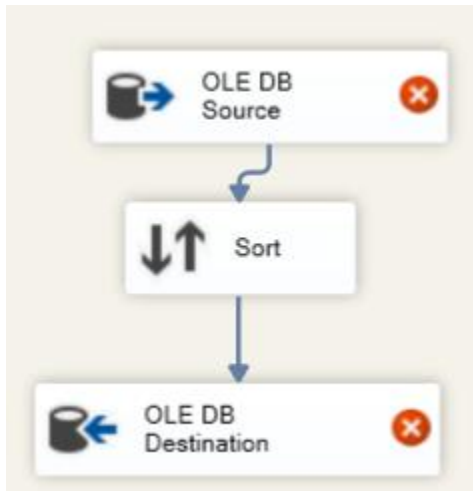
4.1: ETL to Staging Area

Control Flow:



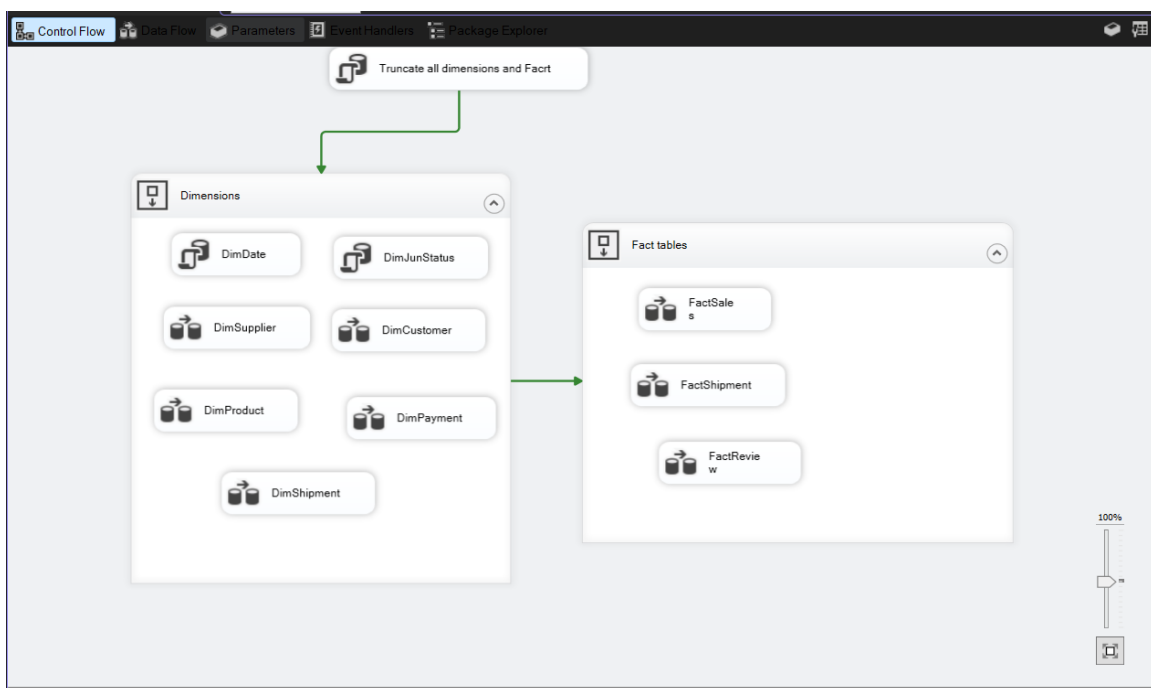


Data Flow:



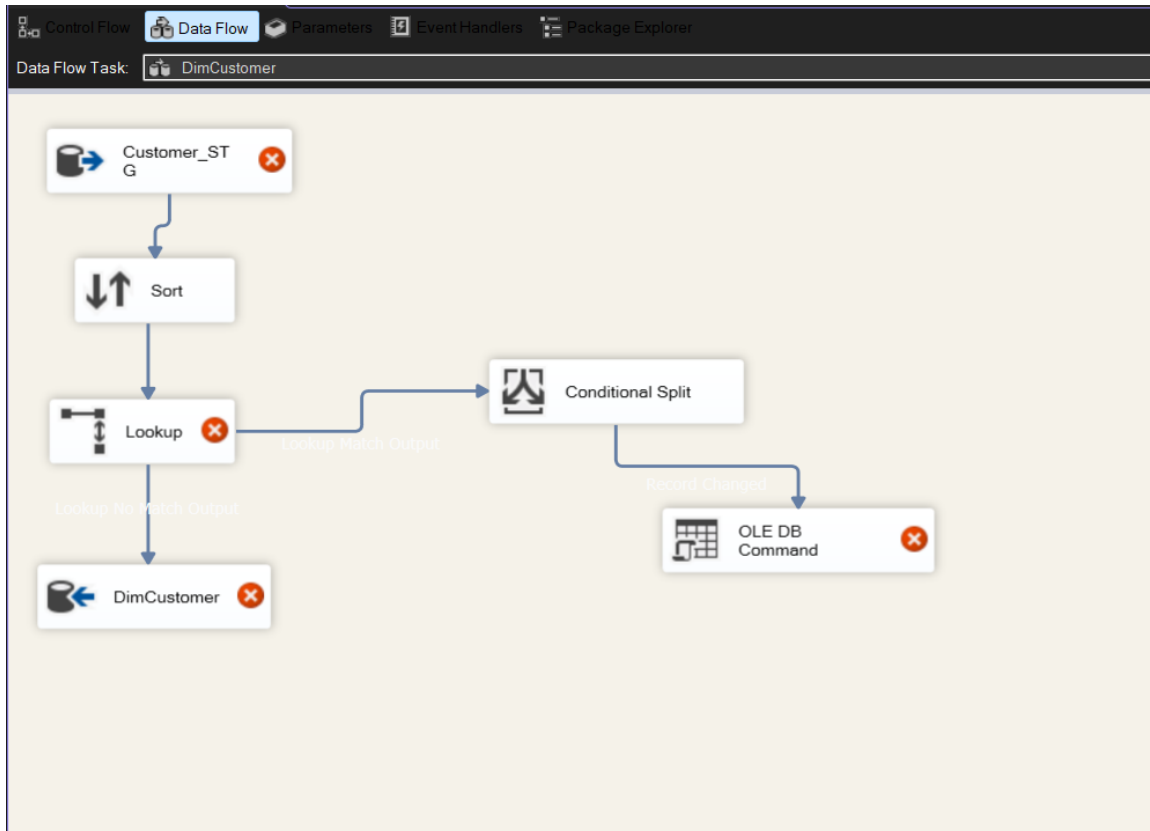
4.2 ETL to DWH

Control Flow





Data Flow





5. Deployment

1. Success of deployment

Start Jobs - MAHMOUD-PC

 **Success** 2 Total
2 Success

Details:

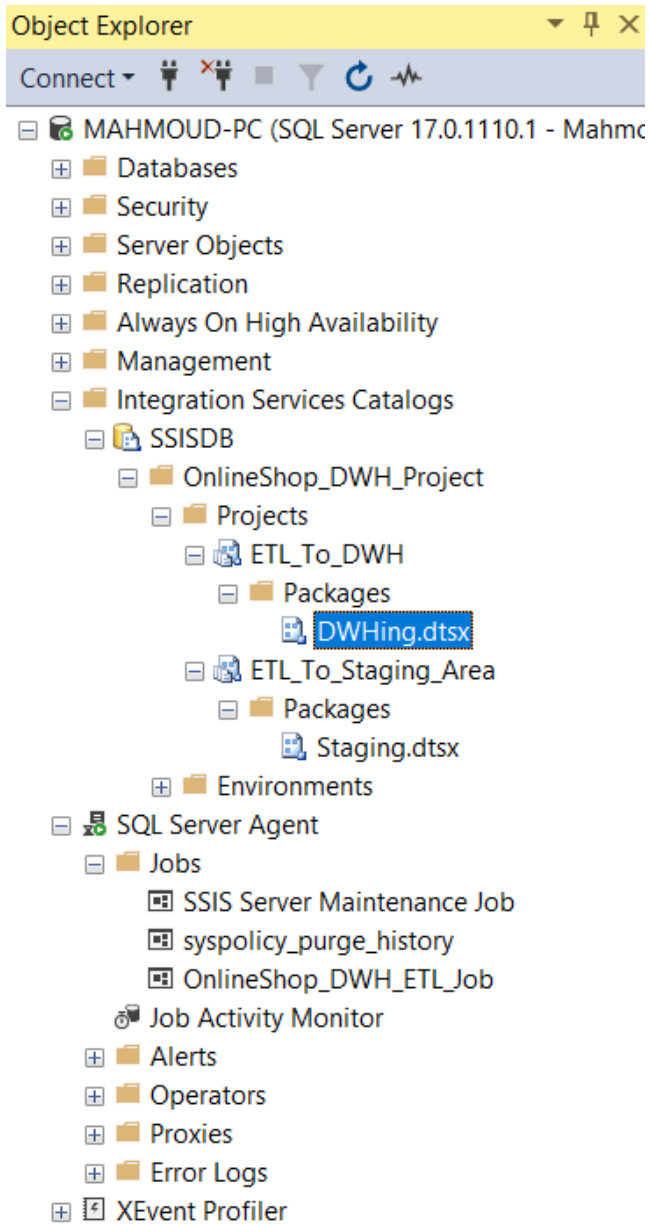
Action	Status	Messa
 Start Job 'OnlineShop_DWH_ETL_Job'	Success	
 Execute job 'OnlineShop_DWH_ETL_Job'	Success	

Close



2. Integration Server Catalogs details in the SSMS explorer.

And showing that all packages (that transfer from the staging to the main DWH) is succeeded





Overview - 11/05/...32 م - Mahmoud-PC

SQL Server

Overview

on Mahmoud-PC at 11/05/2026 05:32:44 م

This report provides an overview of the package tasks and parameters, including execution or validation information.

[View Messages](#)

[View Performance](#)

Execution Information

Operation ID	3
Package	OnlineShop_DWH_Project\ETL_To_Staging_Area\Staging.dtsx
Environment	-
Status	Succeeded
Machine	Mahmoud-PC

Duration (sec)	29.772
Start Time	11/05/2026 05:31:52 م
End Time	11/05/2026 05:32:21 م
Caller	Mahmoud-PC\Mahmo

Parameters Used

Name	Value	Data Type
CALLER_INFO		String
DUMP_EVENT_CODE	0	String
DUMP_ON_ERROR	False	Boolean
DUMP_ON_EVENT	False	Boolean
LOGGING_LEVEL	1	Int32
SYNCHRONIZED	False	Boolean

Execution Overview

[Filter](#) Result All: (3 more)

Result	Duration (sec)	Package Name	Task Name	Execution Path
Succeeded	17.421	Staging.dtsx	Staging	\Staging
Succeeded	16.656	Staging.dtsx	Sequence Container	\Staging\Sequence Container
Succeeded	16.656	Staging.dtsx	Customer	\Staging\Sequence Container\Customer
Succeeded	8.36	Staging.dtsx	Order	\Staging\Sequence Container\Order

Overview - 11/05/...32 م - Mahmoud-PC

Package	OnlineShop_DWH_Project\ETL_To_Staging_Area\Staging.dtsx
Environment	-
Status	Succeeded
Machine	Mahmoud-PC

Start Time	11/05/2026 05:31:52 م
End Time	11/05/2026 05:32:21 م
Caller	Mahmoud-PC\Mahmo

Parameters Used

Name	Value	Data Type
CALLER_INFO		String
DUMP_EVENT_CODE	0	String
DUMP_ON_ERROR	False	Boolean
DUMP_ON_EVENT	False	Boolean
LOGGING_LEVEL	1	Int32
SYNCHRONIZED	False	Boolean

Property Overrides

Property Path	Property Value
---------------	----------------

Execution Overview

[Filter](#) Result All: (3 more)

Result	Duration (sec)	Package Name	Task Name	Execution Path
Succeeded	17.421	Staging.dtsx	Staging	\Staging
Succeeded	16.656	Staging.dtsx	Sequence Container	\Staging\Sequence Container
Succeeded	16.656	Staging.dtsx	Customer	\Staging\Sequence Container\Customer
Succeeded	8.36	Staging.dtsx	Order	\Staging\Sequence Container\Order
Succeeded	8.25	Staging.dtsx	Order Items	\Staging\Sequence Container\Order Items
Succeeded	8.219	Staging.dtsx	Payment	\Staging\Sequence Container\Payment
Succeeded	16.5	Staging.dtsx	Product	\Staging\Sequence Container\Product
Succeeded	8.235	Staging.dtsx	Review	\Staging\Sequence Container\Review
Succeeded	8.219	Staging.dtsx	Shipment	\Staging\Sequence Container\Shipment
Succeeded	8.203	Staging.dtsx	Supplier	\Staging\Sequence Container\Supplier
Succeeded	0.75	Staging.dtsx	Truncate all tables	\Staging\Truncate all tables

1 / 1

15



Overview - 11/05/...36 م - Mahmoud-PC

Overview - 11/05/...32 م - Mahmoud-PC

Overview

on Mahmoud-PC at 11/05/2026 05:36:03 م

This report provides an overview of the package tasks and parameters, including execution or validation information.

[View Messages](#)

[View Performance](#)

Execution Information

Operation ID	4
Package	OnlineShop_DWH\ProjectETL_To_DWH\DW_Hing.dtsx
Environment	-
Status	Succeeded
Machine	Mahmoud-PC

Duration (sec)	11.477
Start Time	11/05/2026 05:34:30 م
End Time	11/05/2026 05:34:41 م
Caller	Mahmoud-PC\Mahmo

Execution Overview

[Filter](#) Result All: (3 more)

Result	Duration (sec)	Package Name	Task Name	Execution Path
Succeeded	9.375	DWHing.dtsx	DWHing	\DWHing
Succeeded	8.515	DWHing.dtsx	Dimensions	\DWHing\Dimensions
Succeeded	8.343	DWHing.dtsx	DimCustomer	\DWHing\Dimensions\DimCustomer
Succeeded	8.328	DWHing.dtsx	DimDate	\DWHing\Dimensions\DimDate
Succeeded	0.046	DWHing.dtsx	DimJunStatus	\DWHing\Dimensions\DimJunStatus
Succeeded	8.515	DWHing.dtsx	DimPayment	\DWHing\Dimensions\DimPayment

Parameters Used

Name	Value	Data Type
CALLER_INFO		String
DUMP_EVENT_CODE	0	String
DUMP_ON_ERROR	False	Boolean
DUMP_ON_EVENT	False	Boolean
LOGGING_LEVEL	1	Int32
SYNCHRONIZED	False	Boolean

Property Overrides

Property Path	Property Value
---------------	----------------

Overview - 11/05/...36 م - Mahmoud-PC

Overview - 11/05/...32 م - Mahmoud-PC

Execution Overview

[Filter](#) Result All: (3 more)

Result	Duration (sec)	Package Name	Task Name	Execution Path
Succeeded	9.375	DWHing.dtsx	DWHing	\DWHing
Succeeded	8.515	DWHing.dtsx	Dimensions	\DWHing\Dimensions
Succeeded	8.343	DWHing.dtsx	DimCustomer	\DWHing\Dimensions\DimCustomer
Succeeded	8.328	DWHing.dtsx	DimDate	\DWHing\Dimensions\DimDate
Succeeded	0.046	DWHing.dtsx	DimJunStatus	\DWHing\Dimensions\DimJunStatus
Succeeded	8.515	DWHing.dtsx	DimPayment	\DWHing\Dimensions\DimPayment
Succeeded	8.484	DWHing.dtsx	DimProduct	\DWHing\Dimensions\DimProduct
Succeeded	8.515	DWHing.dtsx	DimShipment	\DWHing\Dimensions\DimShipment
Succeeded	8.453	DWHing.dtsx	DimSupplier	\DWHing\Dimensions\DimSupplier
Succeeded	0.36	DWHing.dtsx	Fact tables	\DWHing\Fact tables
Succeeded	0.125	DWHing.dtsx	FactReview	\DWHing\Fact tables\FactReview
Succeeded	0.36	DWHing.dtsx	FactSales	\DWHing\Fact tables\FactSales
Succeeded	0.266	DWHing.dtsx	FactShipment	\DWHing\Fact tables\FactShipment
Succeeded	0.469	DWHing.dtsx	Truncate all dimensions and Fact	\DWHing\Truncate all dimensions and Fact

Parameters Used

Name	Value	Data Type
CALLER_INFO		String
DUMP_EVENT_CODE	0	String
DUMP_ON_ERROR	False	Boolean
DUMP_ON_EVENT	False	Boolean
LOGGING_LEVEL	1	Int32
SYNCHRONIZED	False	Boolean

Property Overrides

Property Path	Property Value
---------------	----------------

1 / 1



Some properties when success and failure procedures of main processes [data transfer source -> staging -> DWH]

Job Properties - OnlineShop_DWH_ETL_Job

Select a page

- General
- Steps
- Schedules
- Alerts
- Notifications
- Targets

Script ? Help

Job step list:

S...	Name	Type	On Succ...	On Failure
1	Load Source To Staging	SQL Se...	Go to th...	Quit the...
2	Load Staging To DWH	SQL Se...	Quit the...	Quit the...

Connection

Server:
Mahmoud-PC

Connection:
MAHMOUD-PC\Mahmo

[View connection properties](#)

Progress

Ready

Move step:

Start step:

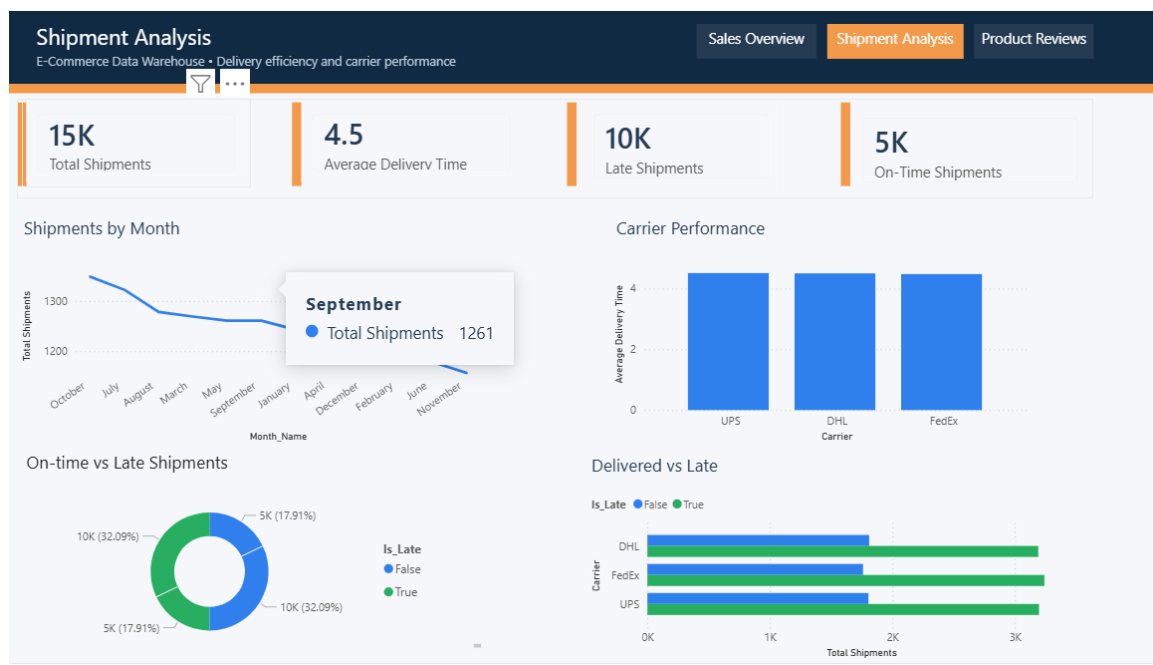
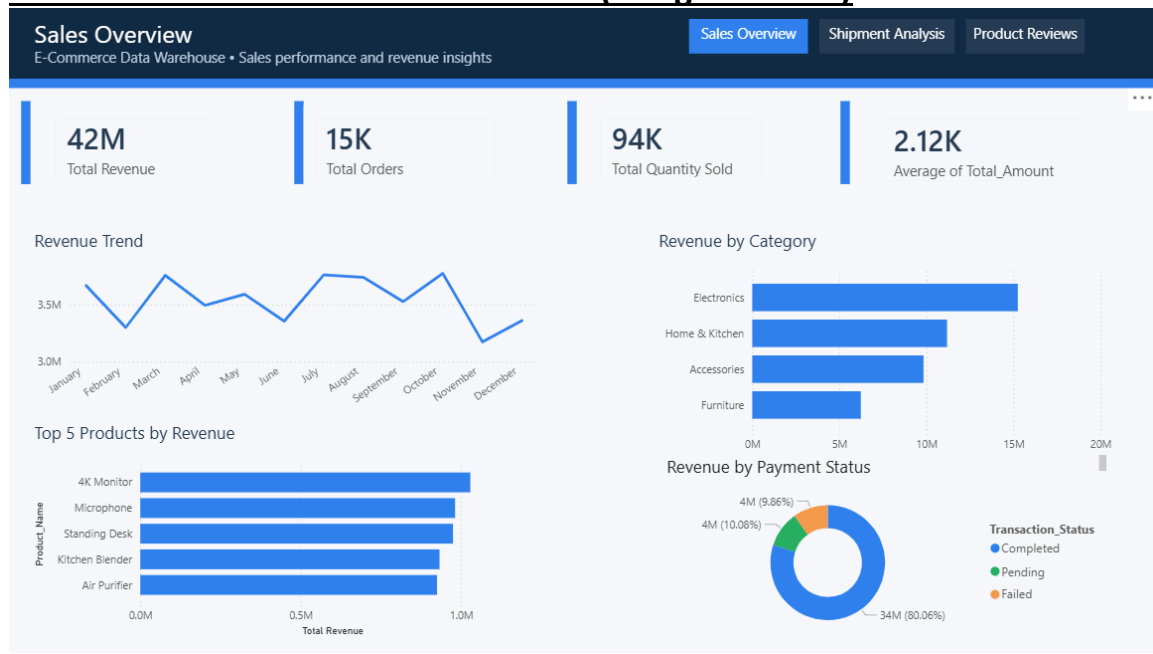
1:Load Source To Staging

New... Insert... Edit Delete

OK Cancel



6. Interactive dashboard for the DWH (using Power BI)





Product Reviews

E-Commerce Data Warehouse • Customer feedback and product quality

Sales Overview

Shipment Analysis

Product Reviews

1K

Total Reviews

2.8

Product Rating Average

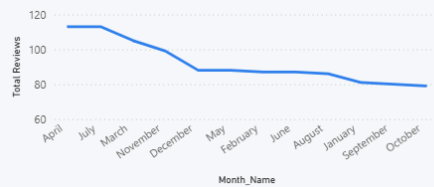
158

Reviewed Products

546

Low Ratings

Reviews Over Time



Average Rating by Category



Top 5 Rated Products



Rating Distribution

